Group II, Row IV: fragmentation function and DGLAP

with the phase $e^{-ik\cdot(y'-y)/\xi}$ kept explicit

In this version the $\frac{1}{\xi}$ of the phase is never absorbed. No "LO at $(\frac{k^+}{\xi}, \frac{\mathbf{k}}{\xi})$" is used. Every equation shows explicitly:

- the phase $e^{-ik\cdot(y'-y)/\xi}$;
- the measure $\frac{d\xi}{\xi^2}$;
- the prefactor $\frac{1}{k^+}$.

Where a parton spectrum is needed at another momentum, the substitution is written out.

1 Your equation

Your ξ form is correct. One change is needed, in blue: the measure is $\int_{x,x',y,y'}$ in both terms. The charge x' of the conjugate amplitude is integrated, as at LO.

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4 d_\perp}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^+\!+\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\xi(1-\xi)+\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}\Big]\frac{1}{\xi^2}\\
&\quad\times\Big[\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{4}\Big]\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x)\\
&\quad+\frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^3}\,\frac{N_c}{k^+}\int_{\frac{k^+}{k^+\!+\Lambda}}^{\frac{k^+}{\vee}}\frac{d\xi}{2\pi}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\,\frac{1}{\xi^2}\Bigg[\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\,d_\perp\,\xi(1-\xi)\\
&\quad+\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Big[\frac{1-\xi}{\xi}+\frac{\xi}{1-\xi}\Big]\Bigg]\big[U^{e'c}(y')-U^{e'c}(x')\big]\big[U^{ec}(y)-U^{ce}(x)\big]\rho^{e'}(x')\rho^{e}(x).\end{aligned}\tag{1.1}$$

It is Eq. (1.5) of `RowIV_dperp.pdf`, term by term (`sympy`: difference 0). The reasons:

- The limits are reversed, $\int_{k^+/(k^+\!+\Lambda)}^{k^+/\vee} = -\int_{k^+/\vee}^{k^+/(k^+\!+\Lambda)}$. This removes the overall minus sign of the E form of (1.5).
- $\frac{1}{\epsilon}+\gamma_E+\frac{1}{2}\log\frac{(y'-y)^2\mu^2}{4} = \frac{E}{2}$, and $\frac{g^4 d_\perp}{4\pi^3}\cdot\frac{1}{2\pi}\cdot\frac{E}{2} = \frac{g^4}{16\pi^4}\,d_\perp E$.
- In the second term, $\frac{g^4}{4\pi^3}\cdot\frac{1}{2\pi} = \frac{g^4}{16\pi^4}\cdot 2$.
- The $\frac{1}{\xi^2}$ and the phase are right. The $\frac{1}{\xi^2}$ comes from $dp^+ = -k^+\frac{d\xi}{\xi^2}$. The phase $\frac{1}{\xi}$ comes from $w'-w = \frac{p^++k^+}{k^+}(y'-y) = \frac{y'-y}{\xi}$, after the w and w' integrations (z drops out of the difference).

Notation used below.

- $\hat\Delta \equiv \frac{y'-y}{|y'-y|}$, $X \equiv \frac{x-y}{(x-y)^2}$, $X' \equiv \frac{x'-y'}{(x'-y')^2}$, and the colour factor (with $U^{bc}=U^{cb}$)
$$\mathcal{O} \equiv \big[U^{ce'}(y')-U^{ce'}(x')\big]\rho^{e'}(x')\,\big[U^{ce}(y)-U^{ce}(x)\big]\rho^{e}(x).\tag{1.2}$$
None of X, X', $\hat\Delta$, $\mathcal{O}$ depends on ξ or $\mathbf{k}$; only the phase does.
- $\xi_0 \equiv \frac{k^+}{\vee}$, $\xi_1 \equiv \frac{k^+}{k^+\!+\Lambda}$, $\delta Y = \log\frac{\vee}{\Lambda}$ and $\delta \equiv \frac{\Lambda}{k^+}$.
- $\frac{1}{\bar\epsilon} \equiv -\frac{2}{\epsilon}$, which is $-\frac{1}{\epsilon_{\rm std}}$ in the usual $D = 4-2\epsilon_{\rm std}$. Then $\frac{E}{2} = -\frac{\bar E}{2}$, with
$$\bar E \equiv \frac{1}{\bar\epsilon}-\ell, \qquad \ell \equiv \log\frac{(y'-y)^2\mu^2_{\overline{MS}}e^{2\gamma_E}}{4} = \log\frac{(y'-y)^2\mu^2_{\overline{MS}}}{c_0^2}, \qquad c_0 \equiv 2e^{-\gamma_E}.\tag{1.3}$$
The pole is collinear (infrared): $\epsilon<0$, $\frac{1}{\bar\epsilon}>0$. $\bar E$ depends on $y'-y$ only, *not* on ξ.
- $P^{(r)}_{gg}(\xi) \equiv 2N_c\big[\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\big] = 2N_c\hat\Phi(\xi)$.

2 Row IV in fragmentation form

Start from Eq. (1.5) of `RowIV_dperp.pdf`:

$$\frac{d^3N_{\rm IV}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^2}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi} \\ \times\Big\{X\cdot X'\Big[d_\perp\hat\Phi(\xi)\bar E - 2\Big(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\Big] - 2d_\perp\xi(1-\xi)(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big\}\mathcal{O}. \tag{2.1}$$

Two identities, neither of which touches the phase or the $\frac{1}{\xi^2}$:

- $\frac{g^4}{16\pi^4} = \frac{\alpha_s}{2\pi}\cdot\frac{g^2}{2\pi^2}$, with $\alpha_s = \frac{g^2}{4\pi}$ (`sympy`: 0);
- $N_c\big[d_\perp\hat\Phi\bar E - 2\big(\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\big)\big] = P_{gg}^{(r)}(\xi)\frac{d_\perp\bar E-2}{2} + 2N_c\xi(1-\xi)$, because $\hat\Phi - \xi(1-\xi) = \frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}$.

Hence, exactly and for any $d_\perp$,

$$\boxed{\begin{aligned}\frac{d^3N_{\rm IV}}{d^3k} &= \frac{\alpha_s}{2\pi}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^2}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi} \\ &\times\Big\{P_{gg}^{(r)}(\xi)\,\frac{d_\perp\bar E-2}{2}\,X\cdot X' + 2N_c\,\xi(1-\xi)\Big[X\cdot X' - d_\perp(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big]\Big\}\mathcal{O}\end{aligned}} \tag{2.2}$$

(`sympy`: braces minus those of (2.1) = 0; minus the compact form of `RowIV_compact.pdf` = 0).

- **The pole.** It sits in the first term only:

$$\left.\frac{d^3N_{\rm IV}}{d^3k}\right|_{\rm pole} = \frac{\alpha_s}{2\pi}\,\frac{d_\perp}{2}\,\frac{1}{\bar\epsilon}\,\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^2}\,P_{gg}^{(r)}(\xi)\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi}\,X\cdot X'\,\mathcal{O}. \tag{2.3}$$

- **The second term has no pole.** $\delta^{ij} - d_\perp\hat\Delta^i\hat\Delta^j$ is traceless in $d_\perp$ dimensions. It is the linearly polarized part of $g\to gg$.
- **The two choices of $d_\perp$.**
  - $d_\perp = 2-\epsilon$: $\frac{d_\perp\bar E-2}{2} = \frac{1}{\bar\epsilon} - \ell + O(\epsilon)$ (`sympy`: 0). The "+1" of $\frac{d_\perp}{2}\frac{1}{\bar\epsilon}$ cancels the -1, so the residue is $P_{gg}^{(r)}$ exactly.
  - $d_\perp = 2$: $\frac{d_\perp\bar E-2}{2} = \frac{1}{\bar\epsilon} - \ell - 1$. The residue is the same; the finite part differs by a scheme change.

3 The endpoint $\xi\to 1$: plus prescription with the phase explicit

Under the ξ integral, $P_{gg}^{(r)}(\xi)$ multiplies $\frac{1}{\xi^2}e^{-ik\cdot(y'-y)/\xi}$. At $\xi=1$ this is $e^{-ik\cdot(y'-y)}$, so the subtraction term of the plus prescription carries the phase $e^{-ik\cdot(y'-y)}$, without $\frac{1}{\xi}$. The plus distribution on $[0,1]$, cut at ξ_0, is $\int_{\xi_0}^1 d\xi\big[\frac{1}{1-\xi}\big]_+ f(\xi) \equiv \int_{\xi_0}^1 d\xi\frac{f(\xi)-f(1)}{1-\xi} + f(1)\log(1-\xi_0)$. With $P_{gg,+}^{(r)}(\xi) \equiv 2N_c\big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\big]$, written out:

$$\int_{\xi_0}^1\frac{d\xi}{\xi^2}P_{gg,+}^{(r)}(\xi)\,e^{-ik\cdot(y'-y)/\xi} \equiv 2N_c\int_{\xi_0}^1 d\xi\Big[\frac{\frac{1}{\xi}e^{-ik\cdot(y'-y)/\xi} - e^{-ik\cdot(y'-y)}}{1-\xi} + \Big(\frac{1-\xi}{\xi^3}+\frac{1-\xi}{\xi}\Big)e^{-ik\cdot(y'-y)/\xi}\Big] \\ + 2N_c\log(1-\xi_0)\,e^{-ik\cdot(y'-y)}. \tag{3.1}$$

Then, exactly (numerically, rel. 7×10^{-14}),

$$\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^2}P_{gg}^{(r)}(\xi)\,e^{-ik\cdot(y'-y)/\xi} = \int_{\xi_0}^1\frac{d\xi}{\xi^2}P_{gg,+}^{(r)}(\xi)\,e^{-ik\cdot(y'-y)/\xi} + 2N_c\log\frac{k^+\!+\Lambda}{\Lambda}\,e^{-ik\cdot(y'-y)} - R_\Lambda(y'-y), \tag{3.2}$$

$$R_\Lambda(y'-y) \equiv \int_{\xi_1}^1 d\xi\Big[\frac{P_{gg}^{(r)}(\xi)}{\xi^2}e^{-ik\cdot(y'-y)/\xi} - \frac{2N_c}{1-\xi}e^{-ik\cdot(y'-y)}\Big] = 2N_c\frac{\Lambda}{k^+}\big[1-ik\cdot(y'-y)\big]e^{-ik\cdot(y'-y)} + O\Big(\frac{\Lambda^2}{k^{+2}}\Big). \tag{3.3}$$

For $k\cdot(y'-y) = 1.3$, numerically $R_\Lambda/\delta = -1.97000 - 2.62408\,i$, $-1.97021 - 2.62276\,i$, $-1.97025 - 2.62263\,i$ for $\delta = 10^{-2}, 10^{-3}, 10^{-4}$, vs. $-1.97025 - 2.62261\,i = 2(1-ia)e^{-ia}$. Here

$$\log\frac{k^+\!+\Lambda}{\Lambda} = \log\frac{\vee}{\Lambda} - \log\frac{\vee}{k^+} + \log\Big(1+\frac{\Lambda}{k^+}\Big). \tag{3.4}$$

The pole of Row IV, exactly.

$$\frac{d^3N_{\rm IV}}{d^3k}\bigg|_{\rm pole} = \frac{\alpha_s}{2\pi}\frac{d_\perp}{2}\frac{1}{\bar\epsilon}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,x',y,y'} X\cdot X'\,\mathcal{O}\bigg\{\int_{\xi_0}^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,e^{-ik\cdot(y'-y)/\xi} + 2N_c\Big[\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}+\log\Big(1+\frac{\Lambda}{k^+}\Big)\Big]e^{-ik\cdot(y'-y)} - R_\Lambda(y'-y)\bigg\}. \tag{3.5}$$

The full Row IV, exactly. $\bar E$ does not depend on ξ, so it comes out of the ξ integral:

$$\frac{d^3N_{\rm IV}}{d^3k} = \frac{\alpha_s}{2\pi}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,x',y,y'}\mathcal{O}\bigg\{\frac{d_\perp\bar E-2}{2}\,X\cdot X'\Big[\int_{\xi_0}^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,e^{-ik\cdot(y'-y)/\xi} + 2N_c\log\frac{k^++\Lambda}{\Lambda}\,e^{-ik\cdot(y'-y)} - R_\Lambda(y'-y)\Big] + 2N_c\int_{\xi_0}^{\xi_1}d\xi\,\frac{1-\xi}{\xi}\,e^{-ik\cdot(y'-y)/\xi}\Big[X\cdot X' - d_\perp(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big]\bigg\} \tag{3.6}$$

(numerically against (2.2), random $d_\perp$, $\bar E$, $X\cdot X'$, $(X\cdot\hat\Delta)(X'\cdot\hat\Delta)$, $k\cdot(y'-y)$, ξ_0, Λ/k^+: rel. 1×10^{-13}).

- The terms with $e^{-ik\cdot(y'-y)/\xi}$ under $\int d\xi$ are the real ($\xi<1$) part of the splitting.
- The terms with $e^{-ik\cdot(y'-y)}$, i.e. without $\frac{1}{\xi}$, are the $\delta(1-\xi)$ terms.
- R_Λ contains $ik\cdot(y'-y)\,e^{-ik\cdot(y'-y)}$. In ξ language this is a $\delta'(1-\xi)$, because $\frac{d}{d\xi}\big[\frac{1}{\xi^2}e^{-ik\cdot(y'-y)/\xi}\big]_{\xi=1} = \big[-2+ik\cdot(y'-y)\big]e^{-ik\cdot(y'-y)}$.

4 The $\overline{\rm MS}$ fragmentation function

Factorization, written out. A final state h with $(k^+,\mathbf{k})$ comes from a gluon with $(q^+,\mathbf{q})$ and momentum fraction ξ:

$$\frac{d^3N_h}{d^3k}(k^+,\mathbf{k}) = \int_{\xi_0}^1 d\xi\,D^{(0)}_{h/g}(\xi)\int dq^+d^2\mathbf{q}\,\delta(k^+-\xi q^+)\,\delta^{(2)}(\mathbf{k}-\xi\mathbf{q})\,\frac{d^3N_g}{d^3q}(q^+,\mathbf{q}). \tag{4.1}$$

The δ functions give $\frac{1}{\xi^3}$ and set $q^+=\frac{k^+}{\xi}$, $\mathbf{q}=\frac{\mathbf{k}}{\xi}$. For the LO gluon spectrum

$$\frac{d^3N_g^{\rm LO}}{d^3q}(q^+,\mathbf{q}) = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2q^+}\int_{x,x',y,y'}e^{-i\mathbf{q}\cdot(y'-y)}\,X\cdot X'\,\mathcal{O}, \tag{4.2}$$

the substitution $q^+=\frac{k^+}{\xi}$, $\mathbf{q}=\frac{\mathbf{k}}{\xi}$ is

- $\frac{1}{q^+}=\frac{\xi}{k^+}$;
- $e^{-i\mathbf{q}\cdot(y'-y)}=e^{-ik\cdot(y'-y)/\xi}$.

So, with $\frac{1}{\xi^3}\cdot\xi=\frac{1}{\xi^2}$,

$$\frac{d^3N_h^{\rm LO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\xi_0}^1\frac{d\xi}{\xi^2}\,D^{(0)}_{h/g}(\xi)\int_{x,x',y,y'}e^{-ik\cdot(y'-y)/\xi}\,X\cdot X'\,\mathcal{O}. \tag{4.3}$$

This has the same structure as the pole of Row IV, Eq. (2.3):

- the same $\frac{d\xi}{\xi^2}$;
- the same phase $e^{-ik\cdot(y'-y)/\xi}$;
- the same $\frac{1}{k^+}$.

The only difference is $D^{(0)}_{h/g}(\xi)\leftrightarrow\frac{\alpha_s}{2\pi}\frac{d_\perp}{2}\frac{1}{\bar\epsilon}P^{(r)}_{gg}(\xi)$. That is why the pole can be absorbed into the fragmentation function.

The pole of the gluon spectrum. Sec. 6 adds the poles of Row IV, of the virtual rows of Group I, and of the charge renormalization of LO. Up to terms of order $\frac{\Lambda}{k^+}$ and $\frac{k^+}{\vee}$, the sum is

$$\frac{d^3N_g}{d^3k}\Big|_{\rm pole}(k^+,\mathbf{k}) = \frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2 k^+}\int_{\xi_0}^1\frac{d\xi}{\xi^2}\,P_{gg}(\xi)\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi}\,X\cdot X'\,\mathcal{O}, \tag{4.4}$$

$$P_{gg}(\xi) = 2N_c\Big[\frac{\xi}{(1-\xi)_+}+\frac{1-\xi}{\xi}+\xi(1-\xi)\Big]+\frac{11N_c}{6}\,\delta(1-\xi)\qquad (n_f=0). \tag{4.5}$$

- The $\delta(1-\xi)$ term gives $\frac{11N_c}{6}e^{-ik\cdot(y'-y)}$.
- Its coefficient is also fixed by momentum conservation. $\int_0^1 d\xi\,\xi\,P^{(r)}_{gg,+} = -\frac{11N_c}{6}$ (`sympy`), so $\int_0^1 d\xi\,\xi P_{gg}=0$ requires $\frac{11N_c}{6}\delta(1-\xi)$.

To use Eq. (4.4) inside Eq. (4.1), it is needed at the gluon momentum $(q^+,\mathbf{q}) = (\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$. Rename its integration variable $\xi\to z$. Then $\frac{1}{k^+}\to\frac{\zeta}{k^+}$, the lower limit $\frac{k^+}{\vee}\to\frac{\xi_0}{\zeta}$, and the phase is

$$e^{-i\mathbf{q}\cdot(y'-y)/z} = e^{-ik\cdot(y'-y)/(\zeta z)}\,: \tag{4.6}$$

$$\frac{d^3N_g}{d^3q}\Big|_{\rm pole}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big) = \frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\frac{1}{(2\pi)^3}\frac{g^2\zeta}{2\pi^2k^+}\int_{\xi_0/\zeta}^1\frac{dz}{z^2}\,P_{gg}(z)\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/(\zeta z)}\,X\cdot X'\,\mathcal{O}. \tag{4.7}$$

The counterterm.

$$\boxed{\Gamma_{gg}(z)\equiv\delta(1-z)-\frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}P_{gg}(z)=\delta(1-z)+\frac{\alpha_s}{2\pi}\frac{2}{\epsilon}P_{gg}(z),\qquad D^{(0)}_{h/g}(\xi)=\int_\xi^1\frac{dz}{z}\,\Gamma_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu\Big).} \tag{4.8}$$

The second form of Γ_{gg} is in your ϵ; $\overline{\rm MS}$ subtracts only $\frac{2}{\epsilon}$. Insert $D^{(0)}$ into Eq. (4.1). At $O(\alpha_s)$ there are two pole terms.

- **A: the counterterm, from LO, Eq. (4.3):**

$$A = -\frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\xi_0}^1\frac{d\xi}{\xi^2}\Big[\int_\xi^1\frac{dz}{z}P_{gg}(z)D_{h/g}\Big(\frac{\xi}{z},\mu\Big)\Big]\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi}\,X\cdot X'\,\mathcal{O}.$$

- **B: the pole of the gluon spectrum, Eq. (4.7), fragmented with** $\int_{\xi_0}^1 d\zeta\,\frac{1}{\zeta^3}D_{h/g}(\zeta,\mu)$ $(\frac{1}{\zeta^3}\cdot\zeta=\frac{1}{\zeta^2})$:

$$B = \frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\xi_0}^1\frac{d\zeta}{\zeta^2}D_{h/g}(\zeta,\mu)\int_{\xi_0/\zeta}^1\frac{dz}{z^2}P_{gg}(z)\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/(\zeta z)}\,X\cdot X'\,\mathcal{O}.$$

In B, change variables $(\zeta,z)\to(\xi,z)$ with $\xi\equiv\zeta z$, at fixed z:

- the phase: $e^{-ik\cdot(y'-y)/(\zeta z)} = e^{-ik\cdot(y'-y)/\xi}$;
- the measure: $d\zeta=\frac{d\xi}{z}$ and $\frac{1}{\zeta^2}=\frac{z^2}{\xi^2}$, so $\frac{d\zeta}{\zeta^2}\frac{dz}{z^2}=\frac{d\xi}{\xi^2}\frac{dz}{z}$;
- the fragmentation function: $D_{h/g}(\zeta)=D_{h/g}(\frac{\xi}{z})$;
- the limits: $\xi_0\le\zeta\le1$, $\frac{\xi_0}{\zeta}\le z\le1$ become $\xi_0\le\xi\le1$, $\xi\le z\le1$.

So $B=-A$ (numerically, with the phase kept: rel. 2×10^{-15}): the counterterm removes the pole, with the same phase $e^{-ik\cdot(y'-y)/\xi}$ on both sides. The kernel $\int_\xi^1\frac{dz}{z}$ arises exactly because $\frac{1}{\zeta z}=\frac{1}{\xi}$ in the phase.

Row IV after the subtraction ($d_\perp=2-\epsilon$). The part of Row IV that remains is Eq. (2.2) without its $\frac{1}{\bar\epsilon}$:

$$\frac{d^3N^{\overline{MS}}_{\rm IV}}{d^3k} = \frac{\alpha_s}{2\pi}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\xi_0}^{\xi_1}\frac{d\xi}{\xi^2}\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi}$$
$$\times\Big\{-P^{(r)}_{gg}(\xi)\log\frac{(y'-y)^2\mu^2}{c_0^2}\,X\cdot X'+2N_c\,\xi(1-\xi)\Big[X\cdot X'-2(X\cdot\hat\Delta)(X'\cdot\hat\Delta)\Big]\Big\}\mathcal{O}. \tag{4.9}$$

The logarithm vanishes at $\mu=c_0/|y'-y|$, i.e. $\mu\sim|\mathbf{k}|$.

5 DGLAP

The cross section through $O(\alpha_s)$, written out.

$$\frac{d^3N_h}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^2(\mu)}{2\pi^2 k^+}\int_{\xi_0}^1 \frac{d\xi}{\xi^2}\, D_{h/g}(\xi,\mu)\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi}\, X\cdot X'\,\mathcal{O} + \int_{\xi_0}^1 \frac{d\zeta}{\zeta^3}\, D_{h/g}(\zeta,\mu)\,\frac{d^3N_{g,\mathrm{NLO}}^{\overline{MS}}}{d^3q}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta};\mu\Big). \tag{5.1}$$

- In $\frac{d^3N_{g,\mathrm{NLO}}^{\overline{MS}}}{d^3q}$, every pole came with its logarithm, $\bar{E} = \frac{1}{\bar{\epsilon}} - \log\frac{L_\perp^2\mu^2}{c_0^2}$, with $L_\perp = y'-y$ (Row IV), $x-w$ (Row II), $y-z$ (Row V 2b). So $\frac{\partial}{\partial\log\mu^2}$ gives -1 times the residues. These are $P_{gg,+}^{(r)}$ plus the $\delta(1-\xi)$ terms of Sec. 6 without the charge renormalization; the soft logarithms and the $\frac{11}{3}$ cancel there.
- $g^2(\mu)$ in the LO term runs, $\frac{d\alpha_s}{d\log\mu^2} = -\frac{\beta_0}{4\pi}\alpha_s^2$. This supplies the $-\frac{\alpha_s}{2\pi}\frac{\beta_0}{2}\delta(1-\xi)$.

Together, at the gluon momentum $(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$:

$$-\frac{\alpha_s}{2\pi}\frac{1}{(2\pi)^3}\frac{g^2\zeta}{2\pi^2k^+}\int_{\xi_0/\zeta}^1\frac{dz}{z^2}\,P_{gg}(z)\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/(\zeta z)}\,X\cdot X'\,\mathcal{O}.$$

Fragmented with $\int\frac{d\zeta}{\zeta^3}D_{h/g}(\zeta,\mu)$ and with $\xi=\zeta z$ as above, $\frac{\partial}{\partial\log\mu^2}\frac{d^3N_h}{d^3k}=0$ reads

$$\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{\xi_0}^1\frac{d\xi}{\xi^2}\Big[\frac{\partial D_{h/g}(\xi,\mu)}{\partial\log\mu^2} - \frac{\alpha_s}{2\pi}\int_\xi^1\frac{dz}{z}P_{gg}(z)D_{h/g}\Big(\frac{\xi}{z},\mu\Big)\Big]\int_{x,x',y,y'} e^{-ik\cdot(y'-y)/\xi}\,X\cdot X'\,\mathcal{O}=0. \tag{5.2}$$

This holds for every k^+, every $\mathbf{k}$, and every projectile and target. Through the phase $e^{-ik\cdot(y'-y)/\xi}$, different ξ are different transverse momenta $\mathbf{k}/\xi$ of the parent. So the bracket vanishes for each ξ:

$$\boxed{\mu^2\frac{\partial}{\partial\mu^2}D_{h/g}(\xi,\mu) = \frac{\alpha_s(\mu)}{2\pi}\int_\xi^1\frac{dz}{z}\,P_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu\Big)} \tag{5.3}$$

with P_{gg} of Eq. (4.5): the time-like DGLAP equation for gluon fragmentation ($n_f=0$).

The same from $D^{(0)}$. In $d_\perp = 2-\epsilon$, g^2 has mass dimension ϵ, so $\frac{d\alpha_s}{d\log\mu^2} = -\frac{\epsilon}{2}\alpha_s + O(\alpha_s^2)$. Then

$$\frac{\partial}{\partial\log\mu^2}\Big[\frac{\alpha_s}{2\pi}\frac{1}{\bar{\epsilon}}\Big] = \frac{\alpha_s}{2\pi}\Big(-\frac{\epsilon}{2}\Big)\Big(-\frac{2}{\epsilon}\Big) = \frac{\alpha_s}{2\pi} \tag{5.4}$$

(`sympy`: ratio 1), and $\frac{dD^{(0)}}{d\log\mu^2}=0$ in Eq. (4.8) gives Eq. (5.3).

Solution to first order.

$$D_{h/g}(\xi,\mu) = D_{h/g}(\xi,\mu_0) + \frac{\alpha_s}{2\pi}\log\frac{\mu^2}{\mu_0^2}\int_\xi^1\frac{dz}{z}P_{gg}(z)\,D_{h/g}\Big(\frac{\xi}{z},\mu_0\Big) + O(\alpha_s^2). \tag{5.5}$$

6 The $\delta(1-\xi)$ part: Row IV and Group I

A $\delta(1-\xi)$ term is one with the phase $e^{-ik\cdot(y'-y)}$, without $\frac{1}{\xi}$. Besides the soft term of Row IV, Eq. (3.5), the poles of this kind are in Group I, Row II and Row V (2b, UV). Their phases are $e^{-ik\cdot(w'-w)}$ and $e^{-ik\cdot(w'-z)}$. Neither has a $\frac{1}{\xi}$; after renaming they are $e^{-ik\cdot(y'-y)}$. With the exact brackets of `Group_I_corrected.pdf` (G5a, G5d):

- **Row II** (per ordering):

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik\cdot(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2(x-w)^2}\,\bar{E}_{x-w}\,B_{\mathrm{II}}\,[U(x')-U(w')][U(w)-U(x)]\rho\rho,$$

$$B_{\mathrm{II}} = \frac{11}{3} + \log\frac{\Lambda^2}{k^{+2}-\Lambda^2} + \log\Big(1-\frac{k^+}{\vee}\Big) - 8\delta + 2\delta^2 - \frac{4}{3}\delta^3.$$

  - With $w\to y$, $w'\to y'$ the colour factor is $-\mathcal{O}$ (numerically: ratio -1).
  - The prefactor is $\frac{\alpha_s}{2\pi}\cdot\frac{N_c}{2}\cdot\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}$.

- **Row V, 2b UV** (per ordering):

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\,B_{\rm V}\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\,E_{y-z}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}[U(x')-U(w')][U(y)-U(z)]\rho\rho,$$

$$B_{\rm V}=2\log\frac{\vee}{\Lambda}-2\log\frac{\vee}{k^+}+2\log\Big(1-\frac{\Lambda}{k^+}\Big)-\frac{11}{6}+4\delta-\delta^2+\frac{2}{3}\delta^3.$$

  - With $y\to x$, $z\to y$, $w'\to y'$ the colour factor is $+\mathcal{O}$ (ratio $+1$).
  - The prefactor is $\frac{\alpha_s}{2\pi}\cdot N_c\cdot\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}$, and $E=-\bar{E}$.

- **Complex conjugate.** Group I is $\mathcal{C}\cdot\mathcal{V}+$ c.c., so both rows count twice. Row IV is $4\mathfrak{C}'^{\dagger}\mathfrak{C}'$ and counts once.

- **Charge renormalization.** $\alpha_0=\alpha_s(\mu)\big[1+\frac{\alpha_s\beta_0}{4\pi}\frac{1}{\bar\epsilon}\big]$ multiplies the LO term by $1+\frac{\alpha_s}{2\pi}\frac{\beta_0}{2}\frac{1}{\bar\epsilon}$.

source	weight	coefficient of $\frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}N_c\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,x',y,y'}e^{-ik\cdot(y'-y)}X\cdot X'\,\mathcal{O}$
Row IV, Eq. (3.5)	1	$2\log\frac{1+\delta}{\delta}=2\delta Y-2\log\frac{\vee}{k^+}+2\log(1+\delta)$; and $-R_\Lambda$
Row II	$\times2$	$-B_{\rm II}=-\frac{11}{3}+2\log\frac{1}{\delta}+\log(1-\delta^2)-\log(1-\xi_0)+8\delta-2\delta^2+\frac{4}{3}\delta^3$
Row V, 2b UV	$\times2$	$-2B_{\rm V}=\frac{11}{3}-4\log\frac{1}{\delta}-4\log(1-\delta)-8\delta+2\delta^2-\frac{4}{3}\delta^3$
charge renormalization		$+\frac{11}{6}$
sum		$\frac{11}{6}+3\log\frac{1+\delta}{1-\delta}-\log(1-\xi_0)$; and $-R_\Lambda$

(`sympy`: sum minus the last line $=0$; $\log\frac{1}{\delta}=\delta Y-\log\frac{\vee}{k^+}$.)

- **The soft logarithms cancel.** δY appears with $+2$ (Row IV), $+2$ (Row II) and -4 (Row V 2b). This needs the weights $1:2:2$, so it also checks the normalization of Row IV against the c.c. in Group I.

- **The $\frac{11}{3}$ and the powers of δ** cancel between Row II and Row V 2b.

- **The $\frac{11N_c}{6}e^{-ik\cdot(y'-y)}$ of P_{gg} is the charge renormalization of LO.** No further row is needed. Together with Row IV this gives Eq. (4.4).

- **Exact remainder.** For finite Λ/k^+ and $k^+/\vee$ the sum leaves

$$\begin{aligned}&\frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,x',y,y'}X\cdot X'\,\mathcal{O}\Big\{N_c\Big[3\log\frac{1+\delta}{1-\delta}-\log\Big(1-\frac{k^+}{\vee}\Big)\Big]e^{-ik\cdot(y'-y)}-R_\Lambda(y'-y)\Big\}\\&=\frac{\alpha_s}{2\pi}\frac{1}{\bar\epsilon}\frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2k^+}\int_{x,x',y,y'}X\cdot X'\,\mathcal{O}\,N_c\Big[4\frac{\Lambda}{k^+}+\frac{k^+}{\vee}+2i\frac{\Lambda}{k^+}\,k\cdot(y'-y)\Big]e^{-ik\cdot(y'-y)}+O\Big(\frac{\Lambda^2}{k^{+2}},\frac{k^{+2}}{\vee^2}\Big),\end{aligned}\qquad(6.1)$$

  using $3\log\frac{1+\delta}{1-\delta}=6\delta+2\delta^3+\ldots$ and Eq. (3.3). The coefficients depend on $\frac{\Lambda}{k^+}$ and $\frac{k^+}{\vee}$, i.e. on the gluon momentum, not on ξ alone. So this cannot be part of $D_{h/g}(\xi,\mu)$. It comes from the different longitudinal limits:

  - $p^+>\Lambda$ in Row IV;
  - $\Lambda<p^+<k^+-\Lambda$ in Row V 2b;
  - the loop up to $\vee$ in Row II.

  It is still open, as already noted in `Part2b_MSbar.pdf`.

7 Checks

Script `frag_checks.py`, results in `frag_checks.json`. All numerical checks keep the phase $e^{-ia/\xi}$ (real and imaginary parts).

check	result
your Eq. (1.1) minus Eq. (1.5) of `RowIV_dperp.pdf` (`sympy`)	0
$\frac{g^4}{16\pi^4} - \frac{\alpha_s}{2\pi}\frac{g^2}{2\pi^2}$ (`sympy`)	0
braces of (2.2) vs. (2.1) and vs. the compact form (`sympy`)	0, 0
$d_\perp = 2-\epsilon$: $\frac{d_\perp \bar{E}-2}{2} - \left(\frac{1}{\bar{\epsilon}} - \ell\right)$ at $O(\epsilon^0)$ (`sympy`)	0
plus prescription with the phase, Eq. (3.2), 24 random cases	rel. 7×10^{-14}
$R_\Lambda/\delta \to 2N_c(1-ia)e^{-ia}$, $a = k\cdot(y'-y)$	see Sec. 3
full Row IV, Eq. (3.6) vs. Eq. (2.2), 12 random cases	rel. 1×10^{-13}
momentum sum rule, $\delta(1-\xi)$ coefficient (`sympy`)	$\frac{11N_c}{6}\delta(1-\xi)$
$B = -A$: change of variables $\xi = \zeta z$ with the phase $e^{-ia/(\zeta z)}$	rel. 2×10^{-15}
$\frac{\partial}{\partial \log \mu^2}\left[\frac{\alpha_s}{2\pi}\frac{1}{\bar{\epsilon}}\right] \Big/ \frac{\alpha_s}{2\pi}$ (`sympy`)	1
colour factors of Row II and Row V 2b relative to $\mathcal{O}$	-1, $+1$
$\delta(1-\xi)$ tally, exact in δ and ξ_0 (`sympy`)	0